WebClient Computing

CSS Basics

Department of Computer Science

Kyungyong Lee

How about Homework

- Homework explanation
- Reference for colors
  - http://www.w3schools.com/colors/colors_names.asp

Fonts

- font-family allows to specify fonts to be used in a document
- Multiple fonts can be written in the priority order in CSV format
  - font-family: Arial, “Times New Roman”, “Courier New”, Serif
- Generic family: A set of fonts with similar look. Has to be used as a last fallback mechanism
  - serif : line at the of characters (명조체)
  - sans-serif : no line at the end of character (고딕체)
  - monospace : all characters have same width (고정폭)
  - cursive (필기체)
  - Fantasy (장식체)

Fonts Family

- Font Family
  - Under the generic family, more fine grained set of fonts
  - Serif – Times New Roman, Georgia
  - Sans-serif – Arial, Vernada
  - Monospace – Courier New, Lucida Console
  - Cursive - Caflisch Script, Corsiva
  - Fantasy - Alpha Geometrique, Critter, Cottonwood, FB Reactor, Studz, Papyrus
- Where to find available fonts
  - http://www.w3schools.com/cssref/css_websafe_fonts.asp
  - http://www.cssfontstack.com/
- Curious about the current fonts in a web page
  - Chrome extension WhatFont

Fonts Example

This font in in Serif generic family

This font in in Sans-Serif generic family

This font in in Monospace generic family

This font in in Cursive generic family

This font in in Fantasy generic family

```html
<!DOCTYPE html>
<html>
<head>
  <title>My first web page</title>
  <style>
    .serif {
      font-family: Times, "Times New Roman", Georgia, serif;
    }
    .sansserif {
      font-family: Verdana, Arial, Helvetica, sans-serif;
    }
    .monospace {
      font-family: "Lucida Console", Courier, monospace;
    }
    .cursive {
      font-family: cursive;
    }
    .fantasy {
      font-family: fantasy;
    }
  </style>
</head>
<body>
  <div class="serif">This font in in Serif generic family</div>
  <div class="sansserif">This font in in Sans-Serif generic family</div>
  <div class="monospace">This font in in Monospace generic family</div>
  <div class="cursive">This font in in Cursive generic family</div>
  <div class="fantasy">This font in in Fantasy generic family</div>
</body>
</html>
```

CSS Position Property

- Position property sets the location method of elements
  - static, relative, fixed
- Position: static
  - The default location (from left and top most)
- Position: relative
  - Position is determined relative to its normal position (static position)
  - top,right, bottom, and left can be set
- Position: fixed
  - Located always at the specified fixed position even with scrolling
  - top,right, bottom, and left can be set

CSS Position Example

This is a relative position

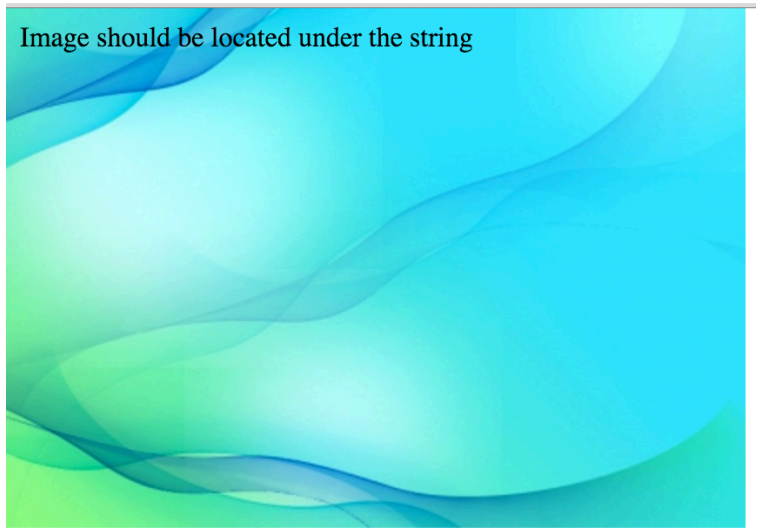
This is a static position

This is a fixed position

```html
<!DOCTYPE html>
<html>
<head>
  <title>My first web page</title>
  <style>
    .static {
      position: static;
      border: 3px solid #73AD21;
    }
    .relative {
      position: relative;
      left: 30px;
      border: 3px solid #73AD21;
    }
    .fixed {
      position: fixed;
      bottom: 0px;
      right: 0px;
      width: 300px;
      border: 3px solid #73AD21;
    }
  </style>
</head>
<body>
  <div class="relative">This is a relative position</div>
  <div class="fixed">This is a fixed position</div>
  <div class="static">This is a static position</div>
</body>
</html>
```

CSS z-index

- It decides whether an element is located on front or back in case of two elements duplicate in a same location
- An element with larger z-index is on top

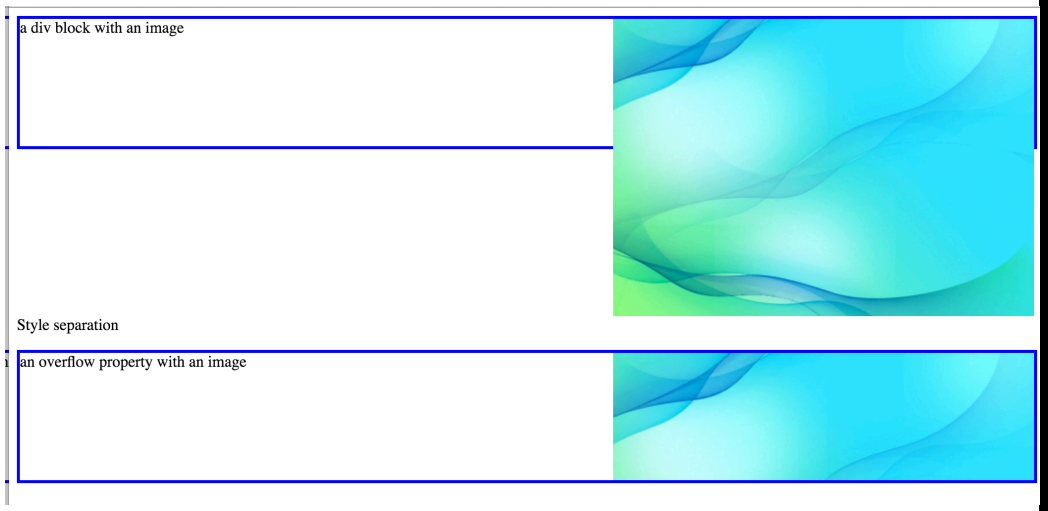



```html
<!DOCTYPE html>
<html>
<head>
  <title>My first web page</title>
  <style>
    .zindex {
      position: fixed;
      left: 0;
      top: 0;
      z-index: -1;
    }
  </style>
</head>
<body>
  Image should be located under the string
  <img class="zindex" src="bg_image.jpg">
</body>
</html>
```

CSS Float and Overflow

- Float property allows to set how an image is located around text
  - left/right
- Overflow property allows to make a containing object to resize based on the size of the internal object
  - auto – automatically set to resize
  - visible – the overflow part is shown (default)
  - hidden – overflow part is not shown
  - scroll – enables a scroll bar for the clipped region

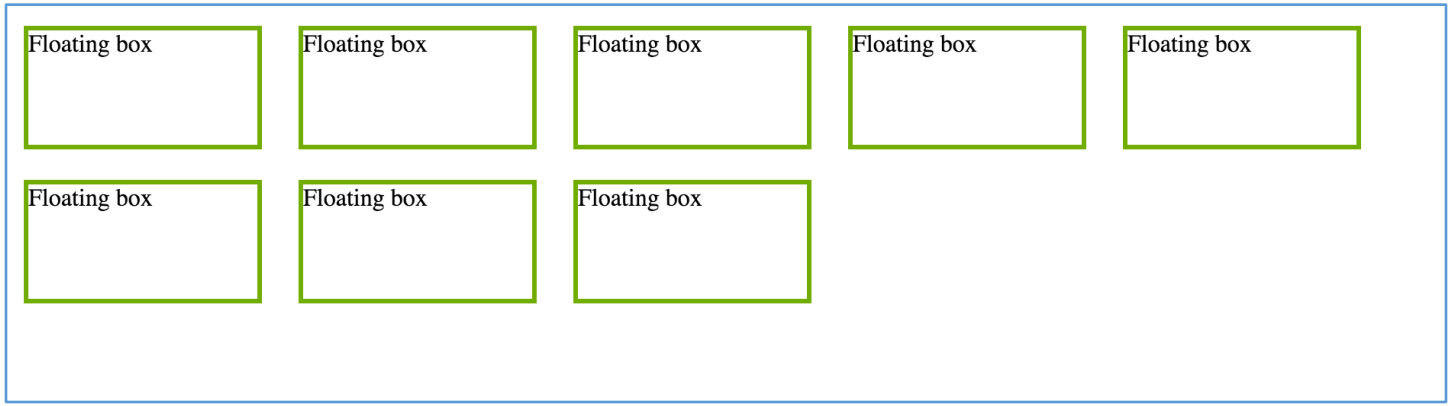
Float Example




```html
<html>
<head>
  <title>My first web page</title>
  <style>
    div {
      border: 3px solid blue;
      width: 1024px;
      height: 128px;
    }
    .float {
      float: right;
    }
    .overflow {
      overflow: scroll;
    }
  </style>
</head>
<body>
  <div>a div block with an image<img class="float" src="bg_image.jpg"></div>
  <p style="clear:right">Style separation</p>
  <div class="overflow">an overflow property with an image
    <img class="float" src="bg_image.jpg"></div>
</body>
</html>
```

Inlineblock

- Inlineblock allows to create a grid of boxes using display property
  - display: inline-block
- Try to resize the page




```html
<!DOCTYPE html>
<html>
<head>
<style>
.floating-box {
    display: inline-block;
    width: 150px;
    height: 75px;
    margin: 10px;
    border: 3px solid #73AD21;
}
</style>
</head>
<body>

<div class="floating-box">Floating box</div>
<div class="floating-box">Floating box</div>
<div class="floating-box">Floating box</div>
<div class="floating-box">Floating box</div>
<div class="floating-box">Floating box</div>
<div class="floating-box">Floating box</div>
<div class="floating-box">Floating box</div>
<div class="floating-box">Floating box</div>

</body>
</html>
```

Combination of CSS Selectors

- Descendant selector - selectorA selectorB
  - The style applies only to parent-child relation in the document
- Child selector – selectorA > selectorB
  - Applies to only immediate children
- Adjacent sibling – selectorA + selectorB
  - Applies to elements that are only adjacent sibling element
- General sibling – selectorA ~ selectorB
  - Applies to all other sibling elements

Combinator Example

Paragraph 1 in the div.

Paragraph 2 in the div.

Paragraph 3 in the div.

Paragraph 4. Not in a div.

Paragraph 5. Not in a div.

descendant

Paragraph 1 in the div.

Paragraph 2 in the div.

Paragraph 3 in the div.

Paragraph 4. Not in a div.

Paragraph 5. Not in a div.

Adjacent sibling (+)

Paragraph 1 in the div.

Paragraph 2 in the div.

Paragraph 3 in the div.

Paragraph 4. Not in a div.

Paragraph 5. Not in a div.

Child(>)

Paragraph 1 in the div.

Paragraph 2 in the div.

Paragraph 3 in the div.

Paragraph 4. Not in a div.

Paragraph 5. Not in a div.

General sibling (~)

```
<!DOCTYPE html>
<html>
<head>
  <title>My first web page</title>
  <style>
    div p {
      background-color: yellow;
    }
  </style>
</head>
<body>
  <div>
    <p>Paragraph 1 in the div.</p>
    <p>Paragraph 2 in the div.</p>
    <span><p>Paragraph 3 in the div.</p></span>
  </div>
  <p>Paragraph 4. Not in a div.</p>
  <p>Paragraph 5. Not in a div.</p>
</body>
</html>
```

References

- http://www.w3schools.com/css/default.asp - CSS Tutorial